National Research University Higher School of Economics (HSE)

Curriculum
Field of study 01.04.02 Applied Mathematics and Informatics
Educational Programme "Math of Machine Learning"
Trajectories: "Math of Machine Learning"
Implementing unit: Faculty of Computer Science, HSE - Moscow
1 st, 2025/2026 academic year

APPROVED
10.04.2025
Vice Rector
ROSHCHIN S.Y.
Signed with EDS

Length of Programme: 2 years
Years of Study: 2025/2026 - 2026/2027
Mode of Study: Full Time
Degree: Master's degree / MBA

Block Code	Course	Subject type	Department	Period of execution	Credits	Total Academic Hours	Contact Hours	Allocation of Contact Hours				Additional Information
								1	2	3	4	
	Degree Programme				**60,00**	**2 280**	**580**	**122**	**162**	**208**	**88**	
	Math of Machine Learning (General track)				**60,00**	**2 280**	**580**	**122**	**162**	**208**	**88**	
	Major				**48,00**	**1 824**	**546**	**112**	**154**	**200**	**80**	
	Basic Components				**27,00**	**1 026**	**314**	**56**	**98**	**120**	**40**	
1	High Dimensional Probability and Statistics	C	Department of Big Data and Information Retrieval		6,00	228	80			40	40A	Foreign language
2	Computational Linear Algebra	C	Skolkovo Institute of Science and Technology		6,00	228	56	28	28A			Foreign language
3	Additional Topics in Probability Theory	C	Department of Big Data and Information Retrieval		3,00	114	28	28A				Foreign language
4	Markov Chains	C	Department of Big Data and Information Retrieval		3,00	114	42		42A			Foreign language
5	Machine Learning	C	Skolkovo Institute of Science and Technology		6,00	228	68		28	40A		Foreign language
6	Sampling and Generative Modeling	C	Department of Big Data and Information Retrieval		3,00	114	40			40A		Foreign language
	Elective Courses. 1 year				**21,00**	**798**	**232**	**56**	**56**	**80**	**40**	
1	Introduction to Data Analysis	E	Skolkovo Institute of Science and Technology		3,00	114	28	28A				Foreign language
2	Introduction to Computer Vision	E	Skolkovo Institute of Science and Technology		3,00	114	28		28A			Foreign language
3	Computational Imaging	E	Skolkovo Institute of Science and Technology		3,00	114	28	28A				Foreign language
4	Geometrical Methods of Machine Learning	E	Skolkovo Institute of Science and Technology		3,00	114	40				40A	Foreign language
5	Deep Learning	E	Skolkovo Institute of Science and Technology		6,00	228	80			40	40A	Foreign language
6	Perception in Robotics	E	Skolkovo Institute of Science and Technology		6,00	228	68		28	40A		Foreign language
7	Deep Learning for Natural Language Processing	E	Skolkovo Institute of Science and Technology		3,00	114	40				40A	Foreign language

8	Optimization in Machine Learning	E	Skolkovo Institute of Science and Technology		3,00	114	28		28A			Foreign language
9	Foundations of Software Engineering	E	Skolkovo Institute of Science and Technology		3,00	114	28	28A				Foreign language
10	Principles of Applied Statistics	E	Skolkovo Institute of Science and Technology		3,00	114	28		28A			Foreign language
11	Tensor Decompositions and Tensor Networks in Artificial Intelligence	E	Skolkovo Institute of Science and Technology		6,00	228	68		28	40A		Foreign language
	Key Seminars				**6,00**	**228**	**32**	**8**	**8**	**8**	**8**	
1	Mentor's Seminar "Math of Machine Learning "	C	Department of Big Data and Information Retrieval		6,00	228	32	8	8	8	8A	Foreign language
	Internship				**6,00**	**228**	**2**	**2**				
	Fixed				**6,00**	**228**	**2**	**2**				
1	Innovation Workshop	C		F	6,00	228	2	2A				Foreign language

Curriculum agreed:

Academic Supervisor	Наумов А.А.	04.04.2025
Dean	ARZHANTSEV I.V.	07.04.2025
Head of Centre for Educational Model Design	LEPESHKIN I.A.	09.04.2025

* Subject type:

Compulsory course	C
Elective course	E

Period of execution:

Fixed	F